Build or Subsidize?

The Welfare Effects of Deployment and Affordability Subsidies in U.S. Broadband

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Motivation: Two barriers to broadband access

- Growing evidence highlights broadband's importance for employment, education, and healthcare
  Atasoy (2013, *ILR Review*); Dettling et al. (2018, *REStat*); Van Parys and Brown (2024, *IJIO*)

Constraints to broadband access

- **Affordability:** service exists, but some households cannot afford it
  15% of broadband users report trouble paying; 34% below $30,000 income (Pew, 2021)
- **Deployment:** high fixed costs leave some locations unserved
  21.3 million Americans lacked 25/3 Mbps fixed broadband at end-2017 (FCC, 2019)

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Two federal programs, Infrastructure Investment and Jobs Act (IIJA, 2021)

- Affordable Connectivity Program (ACP): $14.2B for household broadband bills
- Broadband Equity, Access, and Deployment (BEAD): $42.45B for network deployment

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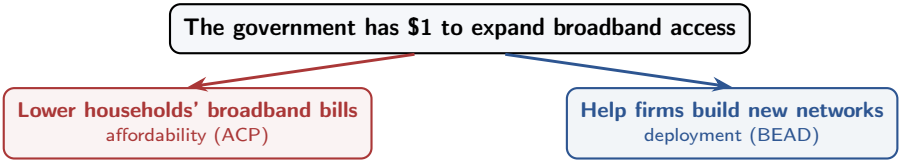
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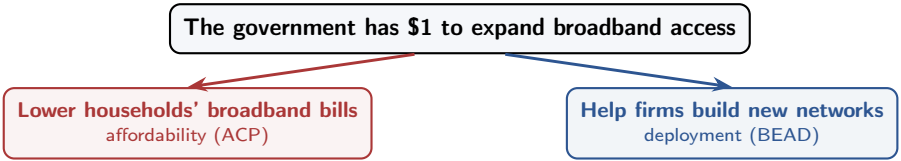
Few local providers: Nearly two-thirds of local areas have at most two providers. Firms' pricing and entry decisions can shape the effects of subsidies

▸ A1: Program timeline

This paper: research question



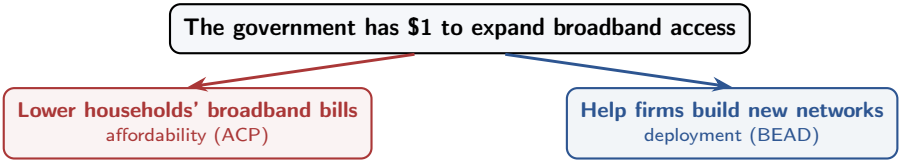
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When should the government subsidize broadband bills, and when should it subsidize network deployment?

The same question arises in electricity, water, transit, and payments

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Approach

1. **Event studies:** examine how markets responded to ACP and BEAD
2. **Structural model:** estimate household demand and model firms' pricing and product choices
3. **Counterfactual analysis:** compare welfare effects per public dollar

Preview of findings

1. **Reduced form: market responses**
   - ACP: lowered posted prices by 7–10% and raised the high-speed share by 6–11pp (P25 vs. P75 exposure)
   - BEAD: limited evidence of early responses in the observed sample
2. **Structural estimates**
   - Consumers respond strongly to prices; typical price–cost margins are around 20%
   - Serving a market requires substantial fixed costs: roughly $\$60,000$˘$\$1$ million per year
3. **Welfare per public dollar**
   - ACP: about \$1 per dollar each year, largely a transfer
   - BEAD: about \$7 per grant dollar over five years (\$1.5 per year)

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Policy implication: the right instrument depends on the local barrier to access

Contributions and related literature

1. **A welfare comparison of affordability and deployment subsidies**

   Compare the two instruments nationally, per public dollar, allowing entry and product choices to respond

   Closest work: Kearns (2026, working paper); Fan and Xiao (2015, RAND); Ackerberg et al. (2014, IIJO)

2. **Fixed-cost bounds with many providers and product choices**

   I adapt profit-bound moment inequalities to 1,732 providers and 70,854 tracts

   Closest work: Ciliberto and Tamer (2009, ECTA); Eizenberg (2014, REStud); Ciliberto et al. (2021, JPE); Fan and Yang (2025, JPE)

3. **Evidence for matching subsidies to local access barriers**

   Deployment gains are largest in monopoly tracts; affordability gains are largest in low-income tracts

   Closest work: Espín and Rojas (2024, IJIO); Kearns (2026, working paper)

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4) Structural Model

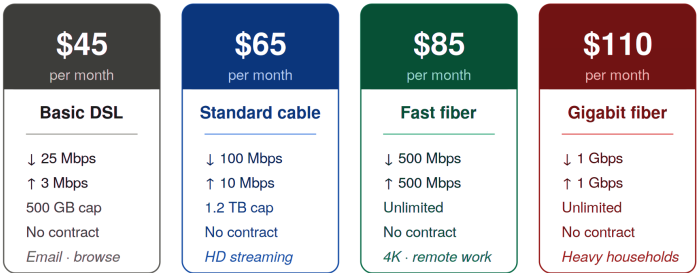
5) Counterfactuals

6) Conclusion

Industry Background and Data

Broadband: differentiated products and local competition

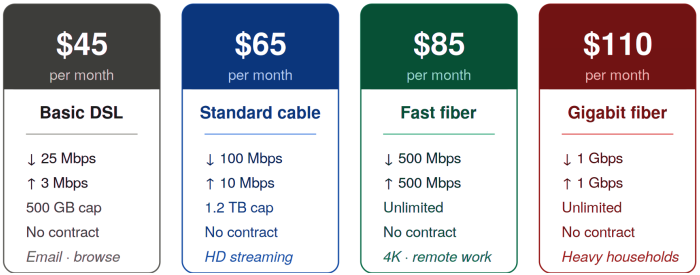
What do households choose? Plans differ in speeds, data allowances, and prices



Illustrative plans

Broadband: differentiated products and local competition

What do households choose? Plans differ in speeds, data allowances, and prices



Illustrative plans

Who provides broadband?

- Telecom and cable operators; some municipal networks
- 1,732 providers; one or two in most tracts

Technologies

DSL	Copper phone lines	**Fiber**	Optical fiber
Cable	Coaxial cable	**Fixed wireless**	Radio links

Data: a new tract-level dataset of U.S. broadband

Sources

- **Federal Communications Commission (FCC) Urban Rate Survey:** plan prices, speeds, technology
- **FCC Form 477:** which providers offer which speed tiers
- **FCC adoption data:** tract-level broadband adoption
- **American Community Survey (ACS), Census Bureau:** income, poverty, households
- **National Telecommunications and Information Administration (NTIA):** BEAD allocations by state

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Two samples

- **Event studies:** state-by-year panel, 52 jurisdictions, 2014–2025 (624 state-years)
- **Structural model:** 727,601 potential provider–speed-tier positions in 70,854 tracts

▸ Descriptive statistics ▸ B1: Sources ▸ B2: Shares ▸ B4: Samples

Descriptive statistics: the typical tract and market structure

The typical tract

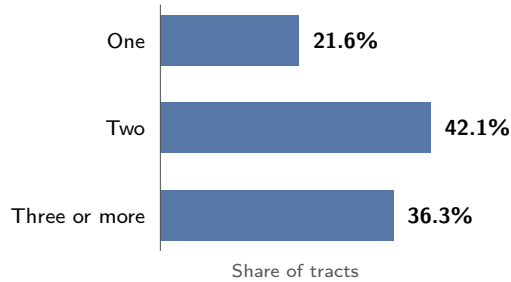
Average monthly price	**$74**
Average population	**4,533**
Median household income	**$65,422**
Households with no broadband	**19.1%**

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Providers per tract



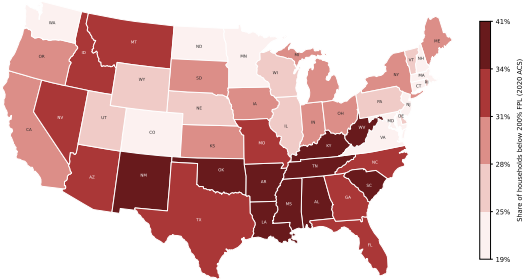
- Median within-tract HHI: **5,472**

Data | B1: Sources | B2: Shares | B3: Coverage

Reduced-Form Evidence

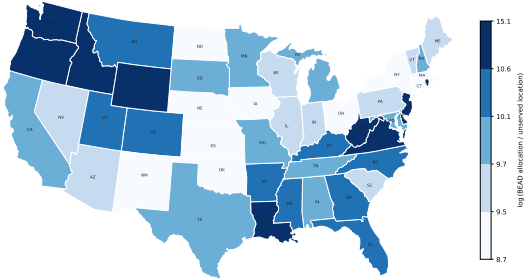
Program exposure varies across states

ACP exposure



Share below 200% Federal Poverty Line (2020 ACS)

BEAD exposure



Log allocation per unserved location

Empirical design: Continuous-treatment event study

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\tau \neq -1} \beta_\tau \left(D_i^P \times 1\{t - \bar{t} = \tau\} \right) + \gamma_i t + \varepsilon_{it}, \quad P \in \{\text{ACP}, \text{BEAD}\}$$

- ACP exposure D_i^{ACP}: 2020 share of households below 200% of the federal poverty line
- BEAD exposure D_i^{BEAD}: log formula allocation per unserved location
- Y_{it}: average price, product mix, or provider HHI in state i
- Event year $\bar{t}$: 2021 (IIJA enactment)
- State and year fixed effects, state trends; weighted by plans; SEs clustered by state

Empirical design: Continuous-treatment event study

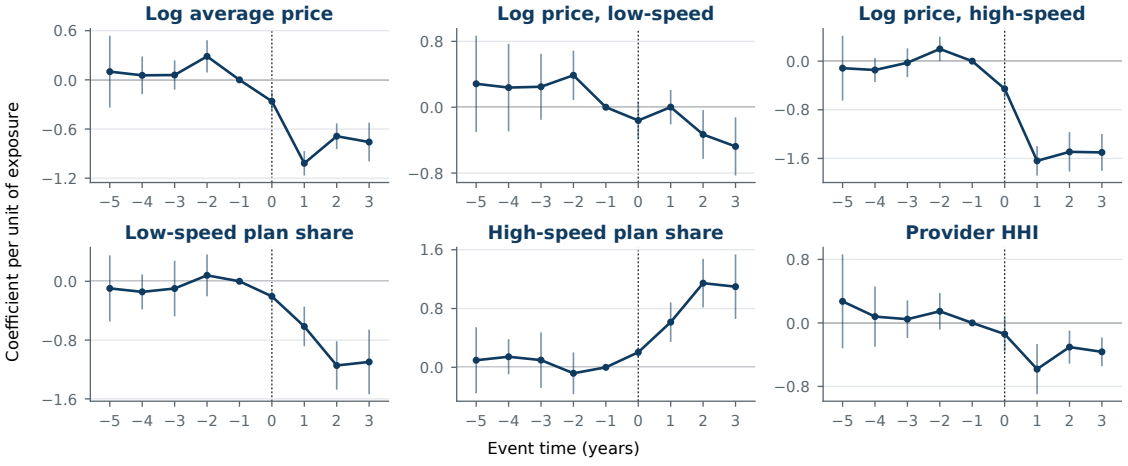
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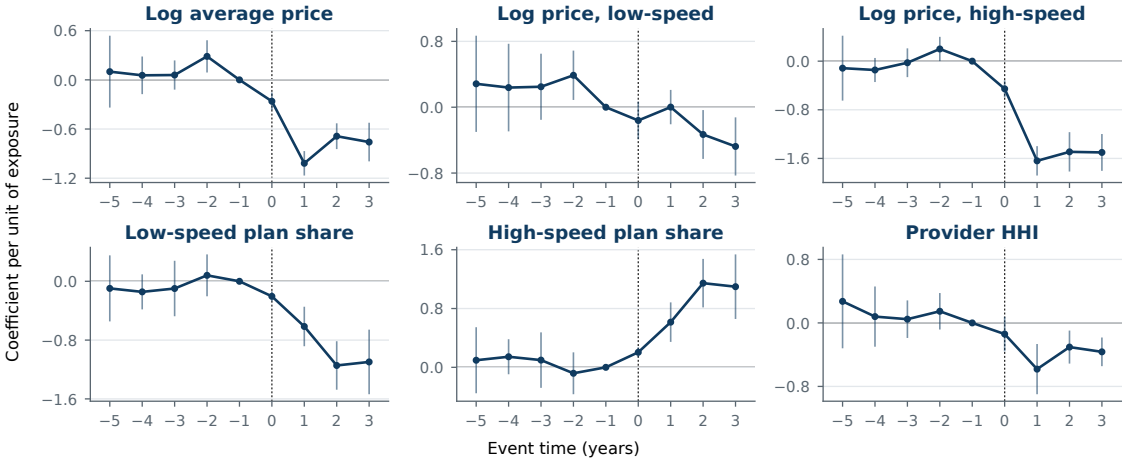
Identification: Absent the programs, outcomes in more- and less-exposed states would have followed similar trends, conditional on fixed effects and state-specific trends

▸ C1: Detail ▸ C2: Anticipation

ACP lowers prices and shifts offerings toward faster plans



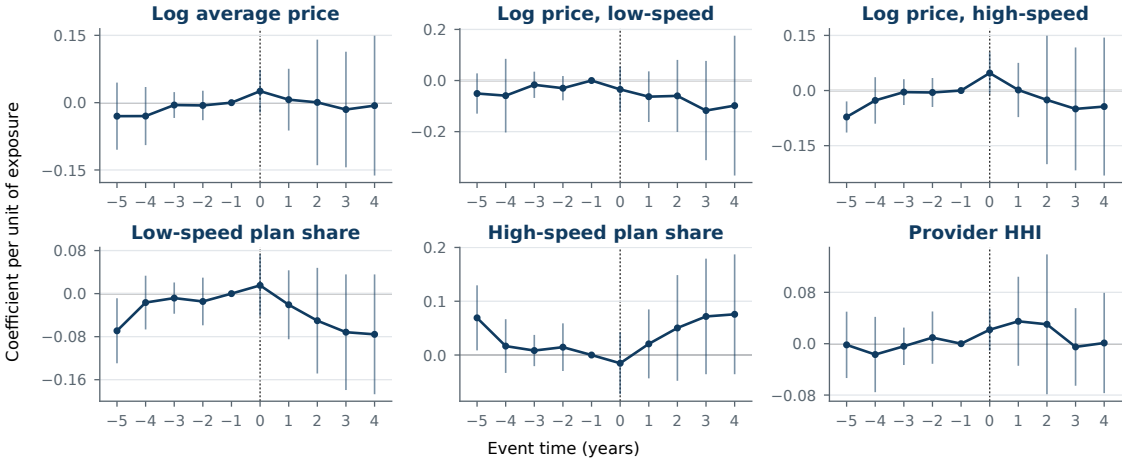
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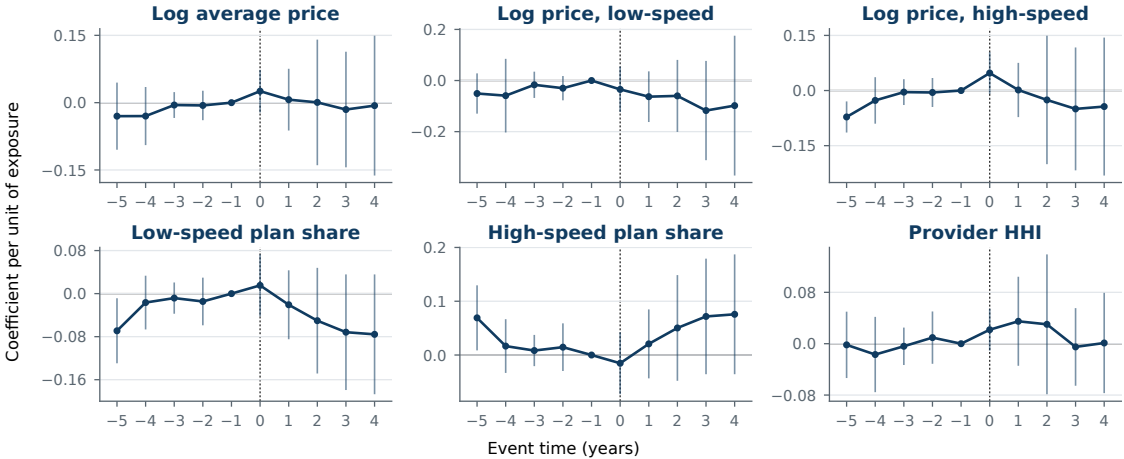
P25 → P75 exposure: price −7 to −10%; high-speed price −15 to −18%; high-speed share +6 to +11pp

▸ C3: Scaled effects ▸ C2: Anticipation ▸ C4: Pooled DiD

BEAD: limited evidence of early responses



BEAD: limited evidence of early responses



Takeaway: limited evidence in the early period; the structural model evaluates deployment subsidies

▸ C4: Pooled DiD

Why do I need a structural model?

The event studies show

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- **Welfare:** who gains, by how much, and per public dollar
- **BEAD once networks are built:** entry where no provider operates today
- **Mechanisms:** new adoption vs. business stealing vs. entry
- **Other designs:** subsidy levels not observed ($10–$30 discounts; 25–75% of fixed costs)

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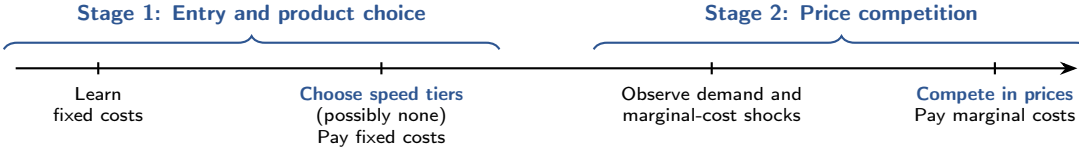
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$\Rightarrow$ **A model of household demand, provider pricing, and entry**, estimated on 70,854 tracts

Structural Model

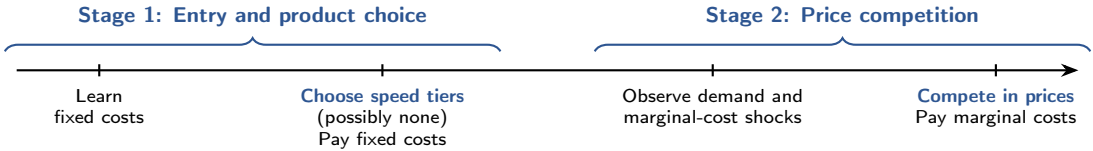
Structural model in a nutshell

Providers play a two-stage game in each census tract



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Estimation: work backward

1. Estimate household demand and recover marginal costs
2. Compute expected variable profits across product portfolios
3. **Multiple entry equilibria** $\Rightarrow$ bound fixed costs (partial identification)

Fan and Yang (2025, JPE); Andrews and Soares (2010, ECTA)

Point-identified: demand, markups. Bounded: fixed costs. Assumed: nested logit, Bertrand-Nash pricing, timing.

▸ Skip to counterfactuals

Second stage: model of household demand for broadband

- **Market:** census tract i
- **Product** j**:** a provider's speed tier (e.g., Verizon, high-speed tier)
- **Nest:** all broadband products in one nest; no broadband is the outside option (nested logit)

Second stage: model of household demand for broadband

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Household h in tract i chooses a product j, or the outside option ($U_{h0i} = \varepsilon_{h0i}$)

$$U_{hji} = \underbrace{X_{ji}^T \beta + \alpha p_{ji} + \xi_{ji}}_{\delta_{ji}\text{: mean utility}} + \nu_{h,\text{in}} + (1-\rho)\,\varepsilon_{hji}$$

- X_{ji}: download and upload speeds; ξ_{ji}: unobserved quality; p_{ji}: monthly price ($100)
- $\nu_{h,\text{in}}$: household taste for broadband, common to all products in the nest
- ε_{hji}: household taste for product j, i.i.d. type I extreme value; $\rho \in [0, 1)$: how alike products are

▸ D1: Why one nest

Second stage: estimation and identification of demand

Estimation: Market-share inversion (Berry, 1994 RAND)

$$\underbrace{\ln s_{ji} - \ln s_{0i}}_{\text{data}} = X_{ji}^{T}\beta + \alpha\, p_{ji} + \rho \ln s_{j|\text{in},i} + \underbrace{\xi_{ji}}_{\text{error}}$$

s_{ji}, s_{0i}: shares of product j and of no broadband; $s_{j|\text{in},i} = s_{ji}/(1 - s_{0i})$: share among broadband adopters

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Endogeneity: high unobserved quality ξ_{ji} raises both p_{ji} and $\ln s_{j|\text{in},i}$ $\Rightarrow$ two endogenous regressors

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Instruments

- Rivals' speed differentiation (Berry et al., 1995 ECTA): excluded from j's utility, shift both p_{ji} and $\ln s_{j|\text{in},i}$
- County and provider fixed effects absorb local demand and brand quality

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- County and provider fixed effects absorb local demand and brand quality

Key assumption (speeds chosen before ξ is realized): $\mathbb{E}\left[\xi_{ji} \mid X_i, Z_i, \text{fixed effects}\right] = 0$

2SLS with county and provider fixed effects; strong first stages (Sanderson–Windmeijer F: 58.3 and 544.2)

▸ D2: Instrument formulas ▸ D3: Instrument strength ▸ D4: Hausman IVs

Second stage: demand estimates

Estimates

- Price sensitivity: $\hat{\alpha} = -1.49$ per \$100
- Product similarity: $\hat{\rho} = 0.86$, so broadband products are close substitutes
- Mean own-price elasticity: -4.99

$\hat{\alpha} < 0$ and $\hat{\rho} \in [0, 1)$, as the model requires

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Own-price elasticities in the literature

- Kearns (2026, working paper), U.S. broadband: -4.73, -4.46, -4.16 for low, middle, high income
- Goetz (2019, working paper), U.S. TV providers: -5.9
- Hidalgo (2023, working paper), Colombia: -4.6
- Crawford et al. (2012, AER), cable and satellite: -4.1 and -5.4

▸ D5: Substitution ▸ D6: Distributions ▸ D7: Specifications

Second stage: pricing reveals markups

Pricing: provider f sets the prices of its products $\mathcal{J}_i^f$ in tract i (Bertrand-Nash)

$$\max_{\{p_{ji}\}_{j\in\mathcal{J}_i^f}} \sum_{j\in\mathcal{J}_i^f} \left(p_{ji} - mc_{ji}\right) M_i \, s_{ji}\left(p_i, X_i, \xi_i\right)$$

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Inversion: stack the first-order conditions and solve for markups

$$p_i - mc_i = -\left(\Omega_i \odot \Lambda_i\right)^{-1} s_i$$

- $\Lambda_{jk,i} = \partial s_{ki} / \partial p_{ji}$, computed from $(\hat{\alpha}, \hat{\rho})$
- $\Omega_{jk,i} = 1$ if j and k belong to the same provider: multiproduct providers internalize cannibalization
- No cost data needed: marginal costs follow as $\widehat{mc}_{ji} = p_{ji} -$ markup

Second stage: marginal costs and markups

Cost function: project recovered marginal costs on product characteristics

$$\log\left(mc_{ji}\right) = X_{ji}^{T}\gamma + \text{fixed effects} + \eta_{ji}$$

Imputes costs for products not offered, needed for Stage 1 portfolio choice

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Results

- Median Lerner index $L_{ji} = (p_{ji} - \widehat{mc}_{ji})/p_{ji}$: 0.20
- Median markup: \$14.3 per month
- Median marginal cost: \$57.5 per month

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- Median markup: \$14.3 per month
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Do these magnitudes make sense?

- Single-product benchmark: $1/|\bar{\varepsilon}| = 1/4.99 \approx 0.20$, the median Lerner index (back of the envelop)
- Implied price \$57.5 + \$14.3 ≈ \$72/month vs. \$71 average U.S. internet bill (Parks Associates, 2024)

▸ D10: Cost estimates ▸ D11: Negative mc ▸ D9: Markups ▸ D6: Distributions

First stage: product positioning: setup and challenge

Setup

- Firms know the second-stage model and the distribution of (ξ, η), not their draws
- Offering product j costs a fixed cost FC_{ji}
- Firms choose portfolios to maximize expected profits net of fixed costs

First stage: product positioning: setup and challenge

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- Multiple equilibria, many firms and products: solving the game is infeasible

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Approach: partial identification

- Undominated choices (Fan and Yang, 2025, JPE) $\Longrightarrow$ moment inequalities
- Inference (Andrews and Soares, 2010, ECTA) $\Longrightarrow$ confidence set

Skip to estimates

First stage: observed product offerings reveal fixed costs

- FC_{ji}: fixed cost of offering product j in tract i (known to firms, not to me)
- Δ_{ji}: expected incremental profit from offering product j, holding the firm's other choices fixed

Offering rule: for an observed offering, $\Delta_{ji} \geq FC_{ji}$

Problem: Δ_{ji} depends on rivals' choices, and equilibria are not unique $\Rightarrow$ bound it (Fan and Yang, 2025 JPE)

$$\underbrace{\Delta_{ji}^{\text{lower}}}_{\text{rivals offer everything}} \leq \Delta_{ji} \leq \underbrace{\Delta_{ji}^{\text{upper}}}_{\text{rivals offer nothing}}$$

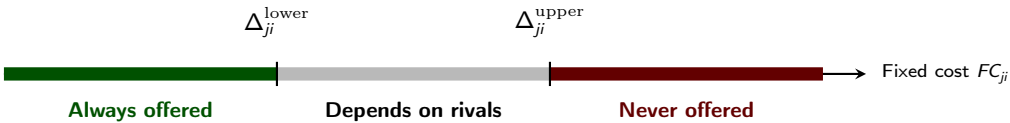
First stage: observed product offerings reveal fixed costs

- FC_{ji}: fixed cost of offering product j in tract i (known to firms, not to me)
- Δ_{ji}: expected incremental profit from offering product j, holding the firm's other choices fixed

Offering rule: for an observed offering, $\Delta_{ji} \geq FC_{ji}$

Problem: Δ_{ji} depends on rivals' choices, and equilibria are not unique $\Rightarrow$ bound it (Fan and Yang, 2025 JPE)

$$\underbrace{\Delta_{ji}^{\text{lower}}}_{\text{rivals offer everything}} \leq \Delta_{ji} \leq \underbrace{\Delta_{ji}^{\text{upper}}}_{\text{rivals offer nothing}}$$



▸ E1: Entry game

First stage: from profit bounds to offer probabilities

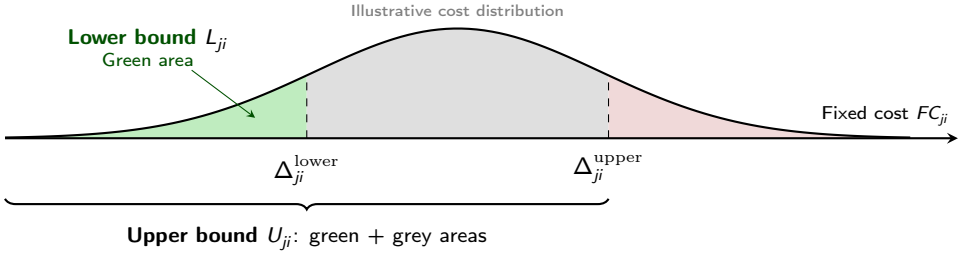
$$FC_{ji} = \mathrm{FC}^{C}_{ji} + \sigma_m \zeta_{ji}, \qquad \zeta_{ji} \sim \mathcal{N}(0,1)$$

$\mathrm{FC}^{C}_{ji}(\theta) = \theta_m + \theta_H 1\{j = H\}$; m: market-size tertile; σ_m: dispersion; F_ζ: standard normal CDF

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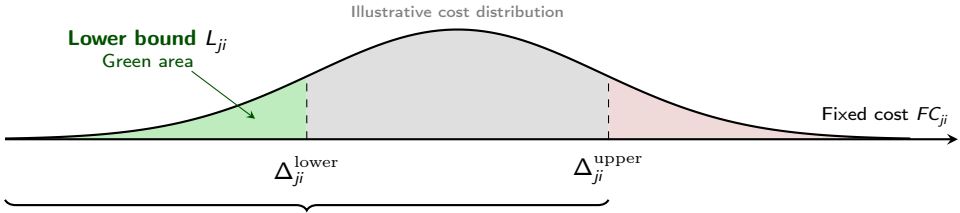
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$$F_\zeta\left(\frac{\Delta_{ji}^{\text{lower}} - \mathrm{FC}^{C}_{ji}(\theta)}{\sigma_m}\right) \leq \Pr(Y_{ji} = 1 \mid X_{ji}) \leq F_\zeta\left(\frac{\Delta_{ji}^{\text{upper}} - \mathrm{FC}^{C}_{ji}(\theta)}{\sigma_m}\right).$$

▸ E2: Bound diagnostics

First stage: moment inequalities for fixed costs

Bounds on offer probabilities $\Rightarrow$ two moment functions

$$m_{ji}^{L}(\theta, \sigma_m) = F_{\zeta}\left(\frac{\Delta_{ji}^{\text{lower}} - \mathsf{FC}_{ji}^{C}(\theta)}{\sigma_m}\right) - 1(Y_{ji} = 1), \qquad m_{ji}^{U}(\theta, \sigma_m) = 1(Y_{ji} = 1) - F_{\zeta}\left(\frac{\Delta_{ji}^{\text{upper}} - \mathsf{FC}_{ji}^{C}(\theta)}{\sigma_m}\right)$$

First stage: moment inequalities for fixed costs

Bounds on offer probabilities ⇒ two moment functions

$$m_{ji}^{L}(\theta,\sigma_m) = F_\zeta\left(\frac{\Delta_{ji}^{\text{lower}} - \mathsf{FC}_{ji}^{C}(\theta)}{\sigma_m}\right) - 1(Y_{ji}=1), \qquad m_{ji}^{U}(\theta,\sigma_m) = 1(Y_{ji}=1) - F_\zeta\left(\frac{\Delta_{ji}^{\text{upper}} - \mathsf{FC}_{ji}^{C}(\theta)}{\sigma_m}\right)$$

Interact with non-negative instruments $g^{(k)}(X_{ji}, W_i)$, $k = 1, \ldots, K$ ($K = 33$: 66 inequalities), and stack:

$$m(\theta,\sigma_m) = \frac{1}{|\mathcal{J}_i|}\sum_{j\in\mathcal{J}_i}\begin{pmatrix} m_{ji}^{L}(\theta,\sigma_m)\, g^{(k)}(X_{ji}, W_i) \\ m_{ji}^{U}(\theta,\sigma_m)\, g^{(k)}(X_{ji}, W_i) \end{pmatrix}_{k=1,\ldots,K} \in \mathbb{R}^{2K}$$

First stage: moment inequalities for fixed costs

Bounds on offer probabilities ⇒ two moment functions

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Identified set: all parameters consistent with every inequality

$$\Theta_0 = \{(\theta,\sigma_m) \in \Theta : \ \mathbb{E}\left[m(\theta,\sigma_m)\right] \leq 0\}$$

First stage: inference on fixed-cost parameters

Test inversion (Andrews and Soares, 2010 ECTA)

- For each candidate (θ, σ_m), test

$$H_0 : \quad \mathbb{E}[m(\theta, \sigma_m)] \leq 0$$

▸ E3: Parameter bounds

First stage: inference on fixed-cost parameters

Test inversion (Andrews and Soares, 2010 ECTA)

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$$H_0 : \quad \mathbb{E}[m(\theta, \sigma_m)] \leq 0$$

- Keep the values that are not rejected:
  $C_n = \{(\theta, \sigma_m) \in \Theta : T_n(\theta, \sigma_m) \leq c_n(0.95; \theta, \sigma_m)\}$

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Search over 7 parameters (a grid is infeasible)

1. Minimize T_n to obtain a starting point.

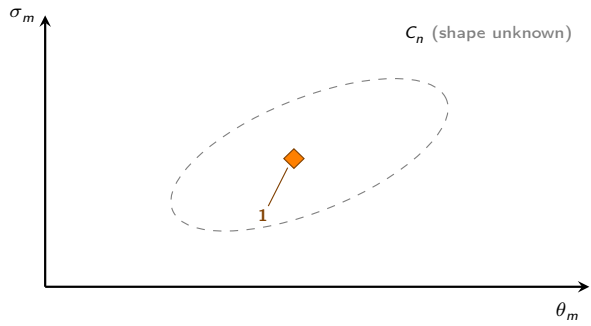


Illustration: two of the seven parameters

▸ E3: Parameter bounds

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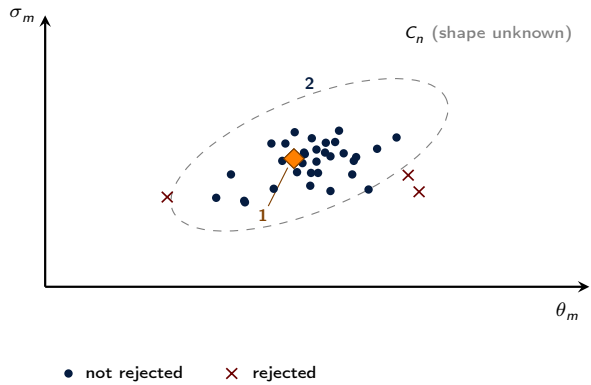


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Search over 7 parameters (a grid is infeasible)

1. Minimize T_n to obtain a starting point.
2. Run random-walk chains; keep draws that are not rejected.
3. Explore further from the edge of the accepted region.

⇒ **656 accepted parameter vectors**

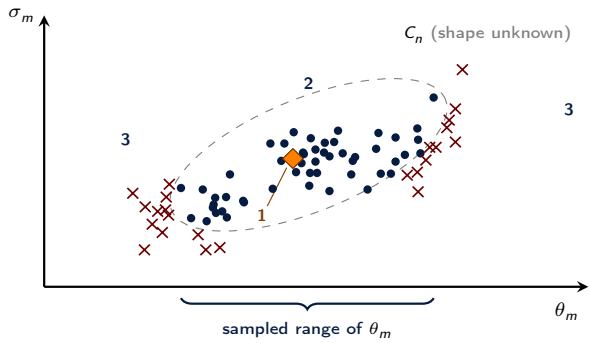


Illustration: two of the seven parameters

▸ E3: Parameter bounds

First stage: fixed costs are large and only partially identified

	95% Confidence Set ($/month)
Mean fixed cost: low-speed position	
Small market	[7,357; 74,287]
Medium market	[5,155; 84,040]
Large market	[5,832; 82,783]
High-speed fixed-cost differential	
High vs. low speed	[−53,843; 25,081]
Fixed-cost heterogeneity	
Small market	[0; 41,542]
Medium market	[0; 45,221]
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- Wide sets reflect **partial identification**
- Mean fixed cost per product: **about $60k–$1M per year**
- Sets overlap across market sizes; dispersion and the high-speed premium are not pinned down
- Same order of magnitude as sunk deployment costs in Kearns (2026, working paper)

▸ E3: Bounds ▸ E4: Plausibility

Counterfactuals

Counterfactuals: affordability versus deployment

Compare each policy with the same **no-subsidy baseline**

ACP: affordability

- Targets **low-income households**
- Up to **$30/month** off broadband service
- Discounts delivered through **participating providers**

In the model: $30 off eligible households' bills; **firms reoptimize prices** and are reimbursed, as under the program

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BEAD: deployment

- Targets **unserved and underserved areas**
- **NTIA** allocates funds to states
- States award **deployment grants** to providers
- Typically up to **75% BEAD funding**, with at least **25% matching funds**

In the model: 25–75% of fixed costs paid; providers re-choose products, then prices

▸ F1: BEAD rules ▸ F2: Accounting

ACP buys affordability; BEAD buys infrastructure

ACP at $30/month

ΔCS per month	$+$\$0.80–0.82B
Public cost	\$0.81–0.83B/month
Discount on gross price	$-$\$30 eligible ($-$\$9 avg.)
Take-up change	$+ \sim$1.5pp
New firm-market entries	10–17

Mechanism: lower household bills $\Longrightarrow$ higher adoption

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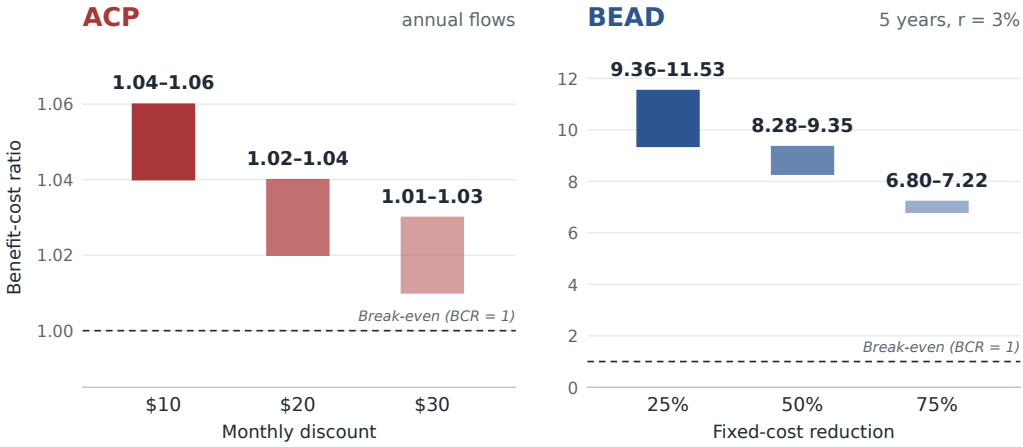
BEAD at 75% fixed-cost subsidy

ΔCS per month	$+$\$1.43–1.84B
Public cost	\$14.3–17.9B one-time
Take-up change	$+\sim$9pp (73% $\rightarrow$ 82%)
New firm-market entries	6,665–8,648
Newly served tracts	6,128–8,054

Mechanism: lower fixed costs $\Longrightarrow$ entry $\Longrightarrow$ coverage

▸ F3: ACP detail ▸ F4: BEAD detail ▸ F5: Repricing

BEAD: $7 per dollar in five years. ACP: near break-even



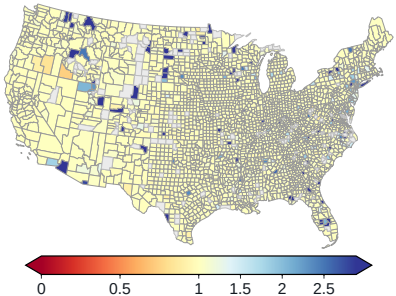
Welfare = ΔCS+ΔPS. ACP: annual welfare / annual outlay, paid every year. BEAD: five-year PV ($r = 3\%$) / one-time grant; per year, about $1.5 per grant dollar.

▸ F7: Sensitivity ▸ F2: Accounting ▸ F8: Sequencing detail

Welfare per public dollar varies across counties

ACP: affordability subsidy

$30/month discount: about **$1** per public dollar



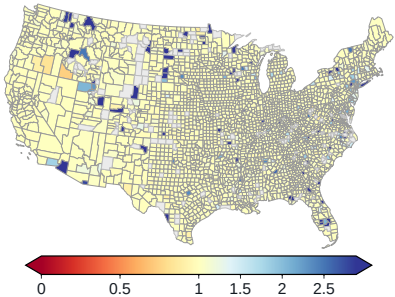
Annual welfare gain / annual spending

$9.9–10.1B / $9.7–9.9B per year

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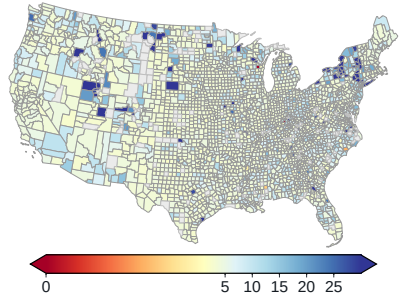
$30/month discount: about **$1** per public dollar



Annual welfare gain / annual spending
$9.9–10.1B / $9.7–9.9B per year

BEAD: deployment subsidy

75% subsidy: about **$7** per public dollar



5-year PV of welfare / one-time grant
$99–129B / $14.3–17.9B ($r = 3\%$)

Colour scales differ across maps. BEAD's cost is treated as a one-time capital grant; ACP's as a recurring transfer.

▸ Returns by subsidy level ▸ F7: Sensitivity

ACP gains track income; BEAD gains track market structure

Consumer-surplus gains by income and market type

Consumer-surplus gain ($/hh/month)	ACP ($30)
Lowest-income quartile	+8.4 to +8.6
Highest-income quartile	+2.2 to +2.4
Monopoly tracts	+3.2 to +3.5
Duopoly tracts	+6.0 to +6.1

ACP gains track income; BEAD gains track market structure

Consumer-surplus gains by income and market type

Consumer-surplus gain ($/hh/month)	ACP ($30)	BEAD (75%)
Lowest-income quartile	+8.4 to +8.6	+16.7 to +20.7
Highest-income quartile	+2.2 to +2.4	+9.9 to +14.4
Monopoly tracts	+3.2 to +3.5	+41.5 to +51.6
Duopoly tracts	+6.0 to +6.1	+2.0 to +3.1

ACP gains track income; BEAD gains track market structure

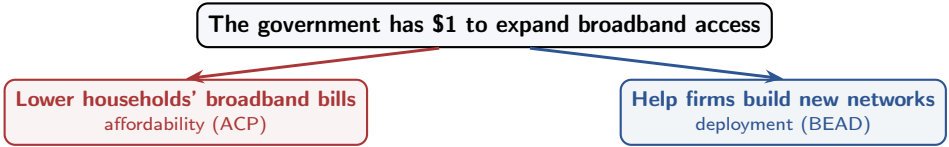
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Monopoly tracts	+3.2 to +3.5	+41.5 to +51.6
Duopoly tracts	+6.0 to +6.1	+2.0 to +3.1

ACP gains follow income (lowest quartile $> 3\times$ highest); BEAD gains follow market structure (monopoly $\gg$ duopoly)

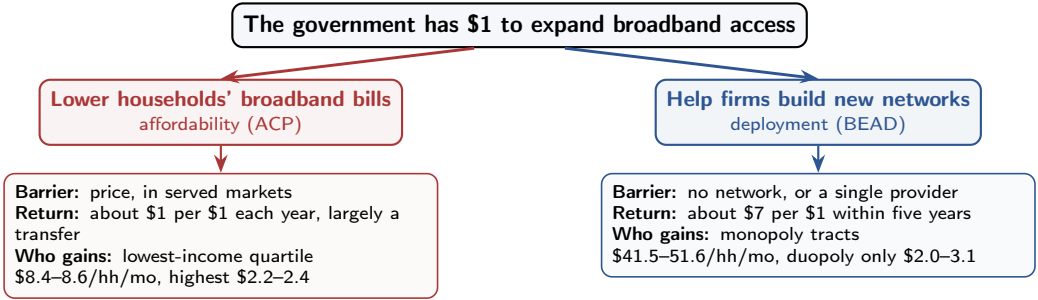
▸ F9: Full table

Policy discussion: back to the $1



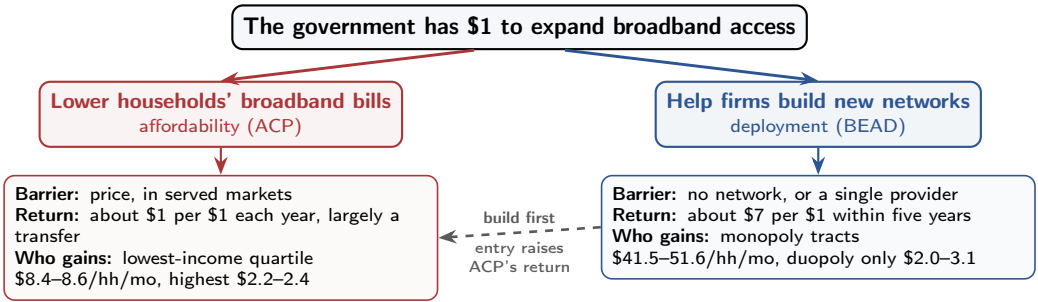
▸ F8: Sequencing detail ▸ F9: Full table

Policy discussion: back to the $1



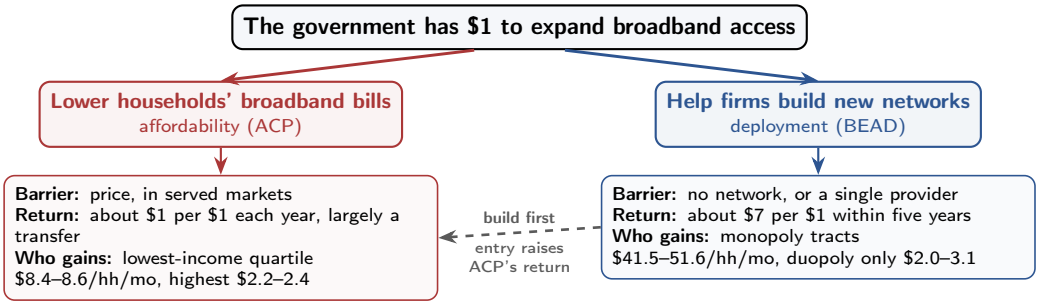
▸ F8: Sequencing detail ▸ F9: Full table

Policy discussion: back to the $1



▸ F8: Sequencing detail ▸ F9: Full table

Policy discussion: back to the $1



Implications for policy design

- **Target by barrier, not by program:** BEAD for unserved and monopoly tracts; ACP by income where networks exist
- **Sequence:** build where networks are missing, then subsidize bills where price still binds
- **Duopoly gap:** ACP reaches these tracts; a subsidized third entrant would need more than 75% of fixed costs

▸ F8: Sequencing detail ▸ F9: Full table

Robustness and validation

Data

Inferred shares: 99.2% of FCC subscriber count — B2

Form 477 overstatement is uniform — B3

Reduced form

Anticipation and policy endogeneity — C2

Pooled DiD confirms event-study dynamics — C4

Urban price frame: BEAD is a lower bound — C6

Demand and pricing

Instrument strength: conditional $F \geq 58$ — D3

Hausman instruments rejected — D4

Nested vs. random-coefficients logit — D1

Inverse-elasticity check on Lerner indices — D9

Negative marginal costs (9.6%) excluded — D11

Fixed costs

Profit bounds: no dominated positions — E2

Magnitudes match Kearns (2026) — E4

Counterfactuals

Bounds across 50 fixed-cost draws — F3 F4

ACP repricing would only add gains — F5

Discount rate and horizon — F7

Scope

External validity — G1

Limitations in full — G2

Conclusion

Conclusion

- **Question:** with one public dollar, lower household bills (ACP) or help firms build networks (BEAD)?
- **Approach:** structural model of household demand, provider pricing, and product offerings, with fixed costs bounded by moment inequalities
- **Findings** (within five years)
  - BEAD returns about **$7 per public dollar**, concentrated in unserved and monopoly tracts
  - ACP returns about **$1 per public dollar**: largely a transfer
  - BEAD gains track market structure; ACP gains track income

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  - BEAD gains track market structure; ACP gains track income
- **Policy implication:** the gains from each instrument depend on the local barrier to access

▸ G2: Limitations

Thank you

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Read the paper

Build or Subsidize? The Welfare Effects of Deployment and Affordability Subsidies in U.S. Broadband

Scan the QR code or click here to access the paper.

Appendix

- A. Introduction
- B. Data
- C. Reduced-form evidence
- D. Demand and pricing
- E. Entry and fixed costs
- F. Counterfactuals
- G. Conclusion

Appendix A1: Program timeline and implementation

- **Feb–Dec 2021**: Emergency Broadband Benefit (EBB), ACP's predecessor, up to $50/month, nearly 9 million households
- **Nov 2021**: IIJA enacted; $14.2B ACP, $42.45B BEAD
- **Jan 2022**: ACP launch; up to $30/month ($75 Tribal), one-time $100 device discount; peak 23 million households
- **Jun 2023**: NTIA announces BEAD state allocations; state proposals due Dec 2023
- **Jun 2024**: ACP funding exhausted; Congress declines to reappropriate
- **2024–2026+**: BEAD subgrantee selection, permitting, construction

ACP's full cycle (launch to lapse) sits inside the 2014–2025 event window; BEAD's does not, which is why its steady state comes from the structural counterfactuals.

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Appendix B1: Data sources

Source	What it contributes	Used for
FCC Urban Rate Survey	product prices, speed tiers, technology	Event studies, demand prices
FCC Form 477	Provider deployment at block/tract level	Product availability, feasible entry set
ACS	Income, poverty, household counts	ACP exposure, market size, controls
FCC adoption data	Tract-level adoption bins	Outside option, market shares
NTIA records	BEAD allocations by state	BEAD treatment intensity

- Event-study panel: 52 states (50 states, DC, Puerto Rico), 2014–2025, $N = 624$ cells
- Demand sample: 170,419 observed products in 66,454 tracts
- Entry sample: 727,601 potential positions in 70,854 tracts

Back

Appendix B2: Market share construction and validation

- Product-level subscriber counts are never disclosed; shares are **inferred**, following the logic of Crawford et al. (2019)
- FCC tract-level adoption bins anchor total adoption and the outside option; logit weights built from product characteristics and demographics allocate the total across available products
- These are inferences, not observed counts, and are treated as such
- **Validation**: the construction implies 93.7M subscribing households at $\geq$10/1 Mbps, 99.2% of the FCC's independent 94.5M benchmark
- Mean outside-option share: 19.1%; mean product share 0.32; provider HHI computed from within-tract subscriber shares

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Appendix B3: Form 477 coverage overstatement

- Known limitation: a block is marked served if a provider *could* serve a household there; this overstates availability in multi-dwelling buildings and rural blocks
- I use the **December 2020** release, before the post-IIJA transition to the Broadband Data Collection system, so the overstatement is **uniform across the cross-section**
- It therefore does not differentially affect the treatment variation used for identification
- Provider normalization (1,732 canonical identifiers) prevents overstating competition by counting subsidiaries separately

Back

Appendix B4: Sample construction

- No single federal source links prices, deployment, demographics, and adoption; five sources are merged at census-tract resolution
- **Brand-first price matching** assigns a price to each of the 727,601 potential positions (provider-tier-tract triples)
- **Provider normalization**: names manually mapped to 1,732 canonical corporate identifiers, so subsidiaries are not counted as separate competitors
- Demand sample (66,454 tracts) is smaller than the entry sample (70,854) because 4,400 tracts with observed products did not match tract-level covariates
- Prices deflated to constant dollars; a product is a provider-speed-tier pair, $g \in \{H, L\}$ split at 10/1 Mbps

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Appendix C1: Event-study specification detail

- Outcomes (six, three margins): log average real price; log price by tier (below/above 25/3 Mbps); product shares by tier; provider HHI
- Prices are product-count-weighted state-year averages, deflated by CPI-U
- Event-time normalization: $\bar{t}^{ACP} = 2022$ (first full implementation year); $\bar{t}^{BEAD} = 2021$ (enactment); omitted period $\tau = -1$
- Five pre-treatment leads test parallel trends; regressions weighted by product counts; SEs clustered at the state level
- **State-specific linear trends**: high-poverty and high-unserved states may sit on different broadband trajectories; trends absorb linear deviations, so identification comes from non-linear departures coinciding with rollout

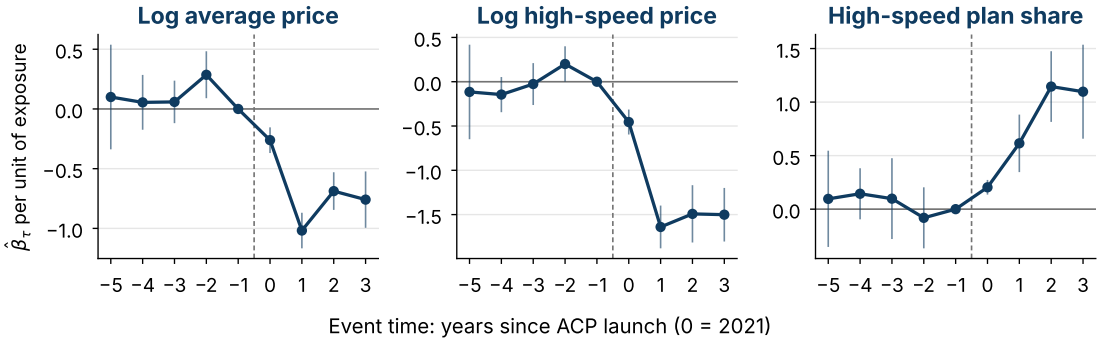
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Appendix C2: Anticipation and policy endogeneity

- **Policy endogeneity**: both intensities are predetermined; 2020 ACS demographics for ACP, a federal formula on pre-grant unserved counts for BEAD; neither responds to post-IIJA ISP behavior
- **BEAD anticipation**: announced Nov 2021, allocations finalized 2023; anticipatory responses would appear at $\tau = 1$–2, where estimates are near zero
- **ACP anticipation**: the EBB ran Feb–Dec 2021, so the $\tau = -1$ base period may absorb some anticipatory ISP behavior; this **understates** ACP's total effect
- Neither concern drives the qualitative results

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Appendix C3: ACP headline effects, scaled to P25 vs. P75 exposure



Effects, P25 vs. P75 exposure

- Prices: -7 to -10% overall, -15 to -18% for high-speed plans
- High-speed plan share: $+6$ to $+11$pp

▸ C4: Pooled DiD ▸ Back

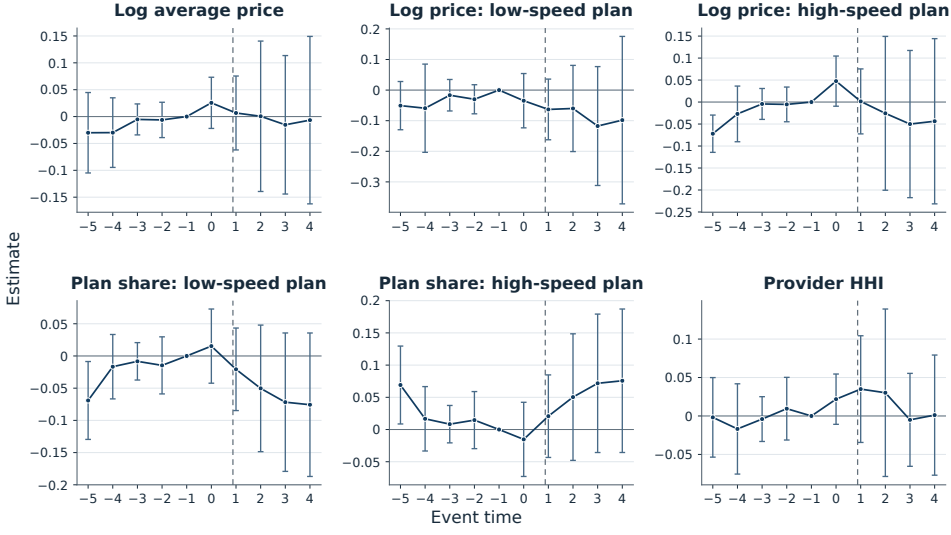
Appendix C4: Pooled difference-in-differences estimates

Outcome	ACP (FPL200)	BEAD (log alloc.)
Log average price	−0.561 (sig.)	small, n.s.
Log high-speed price	−0.922 (sig.)	small, n.s.
High-speed product share	+0.390 (sig.)	small, n.s.
Provider HHI	−0.306 (sig.)	small, n.s.

- Pooled post-period TWFE estimates confirm the event-study dynamics for ACP
- BEAD's pooled estimates are small and insignificant, consistent with effects that emerge only late in the window and that the urban URS frame attenuates
- Flat product-composition results for BEAD are consistent with entry, not incumbent repositioning, as its operative channel

Back to ACP Back to BEAD

Appendix C5: BEAD event-study estimates



Appendix C6: The URS urban frame

- The FCC Urban Rate Survey samples providers in **urban tracts**; the event-study panel under-represents exactly the rural markets BEAD targets
- If BEAD's price and concentration effects concentrate in rural areas, as the structural results imply, URS-based estimates understate its impact: a **conservative lower bound**
- ACP works through demand expansion in already-served, largely urban markets, so the URS frame captures its effects more directly
- This asymmetry is one reason the counterfactuals, not the event studies, carry BEAD's welfare quantification

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Appendix D1: Why nested logit rather than random coefficients

- Random coefficients allow richer substitution but can generate **non-unique pricing equilibria** in multiproduct settings (Nocke and Schutz, 2018): an unstable mapping from primitives to prices
- The counterfactuals **re-solve prices at every iteration** of the positioning game, across 70,854 tracts and 50 draws; a unique closed-form equilibrium is essential
- The single nest applies uniformly: a speed-tier nest is degenerate wherever only one tier is offered (many rural tracts)
- Restrictiveness check: elasticity -4.99 and Lerner 0.20 sit inside the ranges of Kearns (2026, working paper), Goetz (2019), and Crawford et al. (2019)
- Elasticity heterogeneity is still rich through shares: ≈ -2.1 at $s_{jt} = 0.10$ (rural, thin markets) vs. ≈ -8.6 at $s_{jt} = 0.45$ (dense markets); ACP gains concentrate where elasticity is high, BEAD gains where fixed costs bind

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Appendix D2: Differentiation instruments

Four instruments, for each speed characteristic $x \in \{\text{download},\ \text{upload}\}$ (Berry et al., 1995 ECTA):

$$Z_{ji}^{\text{same tier}}(x) = \sum_{\substack{k \neq j \\ \text{same tier}}} (x_{ji} - x_{ki})^2, \qquad Z_{ji}^{\text{other tier}}(x) = \sum_{\substack{k \\ \text{other tier}}} (x_{ji} - x_{ki})^2$$

- Measure how differentiated product j is from rivals' products in the same tract, within and across speed tiers
- Rivals' speeds shift j's price and within-nest share through competition, not through j's utility
- Two endogenous regressors (p_{ji}, $\ln s_{j|\text{in},i}$), four excluded instruments

▸ D3: Instrument strength ▸ Back

Appendix D3: Instrument strength

- Four Berry et al.(1995, ECTA) quadratic differentiation instruments: within- and cross-tier characteristic dispersion in download and upload speed
- Economically transparent signs: more within-tier dispersion raises own price (slack competitive pressure); more cross-tier dispersion lowers the within-nest share
- Sanderson-Windmeijer conditional F: **58.3** (price) and **544.2** (within-share), far above the Stock-Yogo 10% critical value of 7.56
- Cragg-Donald minimum eigenvalue: **986.2**
- Shea partial R^2 for price is 0.0054: the instruments explain a small share of residual price variation, but the identified variation is clean; see D4 for the endogenous alternative

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Appendix D4: Alternative instruments (Hausman-style)

- An alternative specification uses CMT-style Hausman instruments (Ciliberto, Murry and Tamer, 2021): same product's price in other counties
- Stronger first stage on price (Shea $R^2 = 0.393$) but $\hat{\alpha} = -0.123$, close to OLS with fixed effects, and **99.9% of products imply negative marginal costs**
- Convergence to OLS indicates the Hausman instrument is endogenous conditional on two-way fixed effects, likely because national providers carry correlated brand quality across counties
- The (Berry et al., 1995 ECTA) specification is preferred on supply-side plausibility grounds

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Appendix D5: Substitution patterns implied by the nest

Elasticities in closed form from $(\hat{\alpha}, \hat{\rho})$ and observed shares

$$\text{Own price:} \quad \varepsilon_{jj,i} = \hat{\alpha}\, p_{ji} \left[\frac{1}{1-\hat{\rho}} - \frac{\hat{\rho}}{1-\hat{\rho}}\, s_{j|\text{in},i} - s_{ji} \right]$$

$$\text{Cross price } (k \neq j): \quad \varepsilon_{jk,i} = -\hat{\alpha}\, p_{ki} \left[\frac{\hat{\rho}}{1-\hat{\rho}}\, s_{k|\text{in},i} + s_{ki} \right]$$

Appendix D5: Substitution patterns implied by the nest

Elasticities in closed form from $(\hat{\alpha}, \hat{\rho})$ and observed shares

$$\text{Own price:} \quad \varepsilon_{jj,i} = \hat{\alpha}\, p_{ji} \left[\frac{1}{1-\hat{\rho}} - \frac{\hat{\rho}}{1-\hat{\rho}}\, s_{j|\text{in},i} - s_{ji} \right]$$

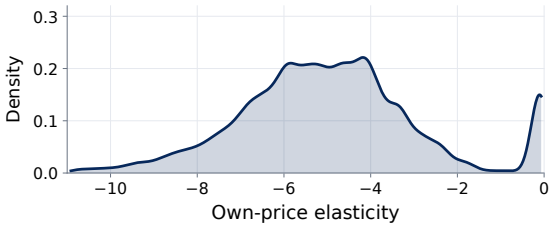
$$\text{Cross price } (k \neq j): \quad \varepsilon_{jk,i} = -\hat{\alpha}\, p_{ki} \left[\frac{\hat{\rho}}{1-\hat{\rho}}\, s_{k|\text{in},i} + s_{ki} \right]$$

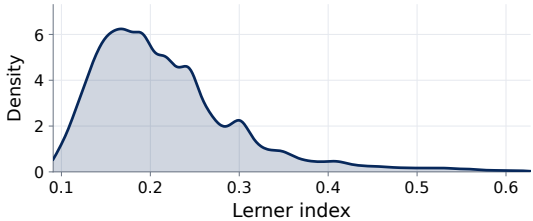
Reading the elasticities

- The within-nest term $\hat{\rho}/(1-\hat{\rho}) \cdot s_{k|\text{in},i}$ captures substitution among broadband products
- With $\hat{\rho} = 0.86$, it carries a weight of about 6: a price cut draws mostly from rival broadband products, much less from non-adopters
- Separates **business stealing** among providers from **new adoption**

▸ D1: Why one nest ▸ Back

Appendix D6: Distributions of elasticities and Lerner indices





Mean own-price elasticity: -4.99

Median Lerner index: 20%

$L_{jt} = (p_{jt} - mc_{jt})/p_{jt}$

Single-product benchmark:

$1/4.99 \approx 20\%$

▸ D8: Demand details ▸ D9: Markups ▸ Back

Appendix D7: Estimation procedure and specifications

Procedure

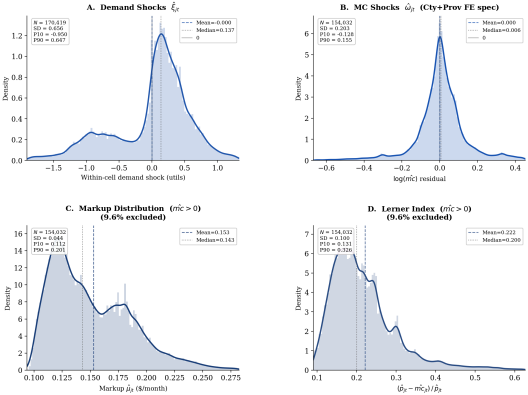
1. Build $\ln s_{ji} - \ln s_{0i}$ and $\ln s_{j|\text{in},i}$ from observed shares
2. Remove county and provider fixed effects (within transformation)
3. 2SLS: instrument p_{ji} and $\ln s_{j|\text{in},i}$ with Z_{ji}; speeds and their squares as controls

	(1) OLS	(2) OLS + fixed effects	(3) 2SLS	(4) 2SLS + fixed effects
Price/\$100 [$\hat{\alpha}$]	-0.344	-0.127	$+3.261$	-1.491
Within-share [$\hat{\rho}$]	1.183	1.029	1.242	0.863
Mean own-price elasticity	$+0.83$	$+1.96$	-5.91	-4.99
County + provider fixed effects	No	Yes	No	Yes

- Only (4) gives $\hat{\alpha} < 0$ and $\hat{\rho} \in [0, 1)$: without IVs, $\hat{\rho} > 1$; without fixed effects, the price coefficient flips sign
- $N = 170{,}419$; state-level aggregation gives elasticity -5.87

▸ D3: Instrument strength ▸ Back

Appendix D8: Demand and supply diagnostics



- Demand shocks $\hat{\xi}_{jt}$ centered near zero: instruments orthogonal to the demand residual after county and provider absorption
- Marginal-cost shocks likewise symmetric
- Markup distribution right-skewed, driven by multiproduct firms
- Lerner median 0.200 matches the inverse-elasticity benchmark 1/4.99

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Appendix D9: Markups and Lerner indices by tier and firm type

	All	Tier H	Tier L
Median markup ($/month)	14.3	14.2	15.3
Median marginal cost ($/month)	57.5	58.9	43.0
Median Lerner index	0.200	0.195	0.278
		Single-product	Multi-product
Median Lerner index		0.191	0.278
Mean Lerner index		0.204	0.322

- Single-product mean Lerner 0.204 $\approx 1/|\bar{\varepsilon}| = 0.200$: the inverse-elasticity check passes
- Tier L (legacy DSL) marks up more: de facto monopolists in low-density areas
- Multiproduct firms internalize cannibalization and price both tiers higher: a new rival disciplines prices in a way an incumbent's second tier does not, the margin on which BEAD operates

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Appendix D10: Marginal-cost estimates

- Recovered costs regressed on characteristics: $\ln(\widehat{mc}_{jt}) = W'_{jt}\gamma + \omega_{jt}$, specification without fixed effects preferred because cost variation runs across providers, technologies, and tiers
- Baseline variable cost $\approx$ **\$33/month** at zero measured speeds; Tier H premium $\approx$ **70%**
- Mean marginal cost $\approx$ \$59.5/month; Tier H \$61.5 vs. Tier L \$38.4: faster connections require heavier plant
- Higher upload speeds correlate with *lower* costs, consistent with fiber-grade networks combining upload capacity with newer, cost-efficient plant
- Download speed raises cost at a diminishing rate
- Column (1) imputes marginal costs for all 727,601 potential positions in the entry sample

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Appendix D11: Negative marginal-cost diagnostics

- 16,387 of 170,419 products (**9.62%**) imply negative marginal costs
- A known artifact of Bertrand-Nash markup recovery: cannibalization-adjusted markups can exceed price when substitution within a firm's own portfolio is strong
- Negative-mc products cluster among **multiproduct firms** (13.5% vs. 8.9% for single-product firms)
- Cost regressions and entry-stage imputation use the positive-mc subsample (90.4%)
- The preferred specification minimizes the problem: the Hausman alternative (D4) produces 99.9% negative costs and is rejected on exactly these grounds

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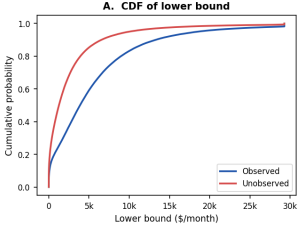
Appendix E1: The entry game

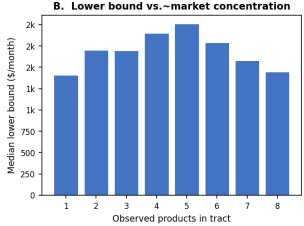
$$\max_{Y_t^f \in \mathcal{Y}_t^f} \mathbb{E}_{\xi,\eta}\left[r_t^f(Y_t^f, Y_t^{-f}) \mid \mathcal{I}^f\right] - \sum_{j \in \mathcal{J}_t^f} Y_{jt}\left(\theta_{b(t)} + \theta_H 1\{g = H\} + \sigma_{b(t)}\zeta_{jt}\right)$$

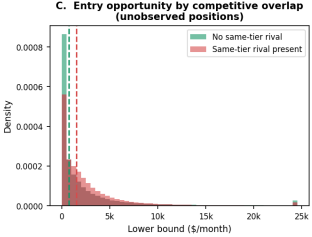
- Firms choose product portfolios before demand and cost shocks realize; expected variable profits come from the estimated demand and pricing stage
- Multiple equilibria imply **bounds** rather than point estimates; no selection rule is imposed
- Moment inequalities compare observed choices with one-position deviations: drop an offered product, or add a feasible unoffered one
- Identification: cross-tertile offering rates discipline θ_b; within-tertile H vs. L comparisons discipline θ_H; the sharpness of the entry threshold disciplines σ_b
- BEAD enters as a proportional fixed-cost reduction

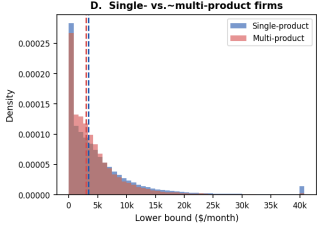
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Appendix E2: Profit-bound diagnostics









- Median tightness ratio $\Delta^{lower}/\Delta^{upper}$ is 0.017: competitive lower bounds are small relative to monopoly upper bounds
- Observed positions have systematically higher ratios: firms select into profitable opportunities
- All valid positions satisfy $\Delta^{lower} > 0$ and $\Delta^{lower} \leq \Delta^{upper}$: no dominated positions; the upper-bound restriction drives identification

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Appendix E3: Fixed-cost parameter bounds

	95% CS ($/month)
θ_H (high-speed premium)	[−53,843; 25,081]
θ_{Small}	[7,357; 74,287]
θ_{Medium}	[5,155; 84,040]
θ_{Large}	[5,832; 82,783]
σ_{Small}	[0; 41,542]
σ_{Medium}	[0; 45,221]
σ_{Large}	[0; 37,891]

- Coordinate-wise projections of the 95% Andrews-Soares confidence set; intercepts by market-size tertile
- Intervals reflect partial identification under multiple equilibria, not sampling noise
- Counterfactuals draw 50 parameter vectors from the set and report outcome bounds

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Appendix E4: Are the fixed-cost magnitudes plausible?

- Monthly intercept bounds of \$5,155–\$84,040 correspond to roughly **\$62k–\$1.0M per year** per product position: the flow equivalent of annualized capital and operating expenditure
- Kearns (2026, working paper) recovers sunk deployment costs of \$560k–\$770k per location for de novo entry (DSL to fiber) from revealed decisions in Seattle: the same order of magnitude
- Revealed-preference cost estimates exceed engineering figures because they embed opportunity costs and regulatory barriers; that applies equally here
- σ_b lower bounds touch zero: a nearly deterministic fixed-cost structure is not ruled out
- θ_H straddles zero: with construction sunk, ongoing fixed costs of a fast tier need not exceed a slow tier's

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Appendix F1: BEAD allocation formula and eligibility

- A federal formula allocates BEAD funds to states by the count of **unserved and underserved locations**: broadband-serviceable locations with no service, service below 25/3 Mbps, or latency above 100 ms
- Counts are measured **pre-grant**: treatment intensity is fixed before any post-IIJA outcome
- BEAD can cover up to **75% of eligible project costs**, with a minimum 25% subgrantee match; this motivates the 25/50/75% counterfactual grid
- Exposure measure: log allocation per unserved location; higher values mean a more intensive supply-side intervention per targeted location

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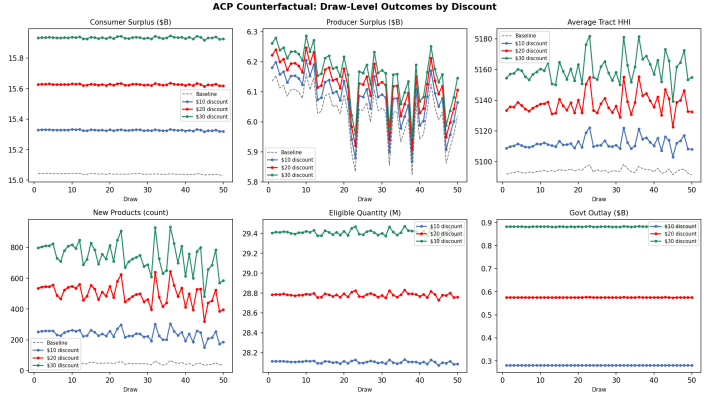
Appendix F2: Welfare accounting and CBA conventions

	BEAD	ACP
Fiscal cost	One-time grant lowering private fixed costs	Recurring transfer on eligible subscriptions
Welfare gain	PV of annual ΔCS + ΔPS, $r = 3\%$, $T = 5$	Annual flow ΔCS + ΔPS
Statistic	NPV and BCR	Annual net surplus and BCR

- Benefits are the discounted surplus the model attributes to each program, **nothing else**: no spillovers into employment, schooling, or health, so ratios understate full social returns
- The model delivers an annual flow, not a time path, so any horizon is an assumption laid on top of it. A five-year window makes that assumption cheap and is the **most conservative** horizon in the sensitivity table
- The two BCRs are not directly comparable levels: one prices a capital stock, the other a flow; the object of interest is the **sequencing** they jointly imply
- Estimates exclude administrative and implementation costs on both sides

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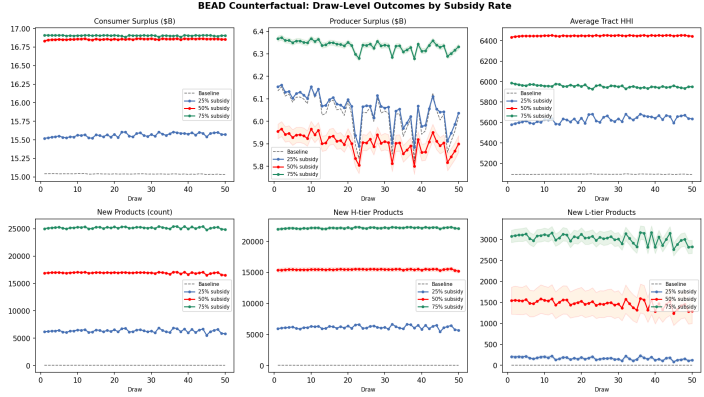
Appendix F3: ACP draw-level outcomes



ACP outcomes are stable across draws because the operative channel is demand rather than entry; fixed-cost uncertainty barely matters for a price discount.

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Appendix F4: BEAD draw-level outcomes



Across 50 fixed-cost draws, BEAD generates large new-product responses, especially at the 75% rate. Reported bounds are $[Q_{2.5}, Q_{97.5}]$ across draws combined with equilibrium multiplicity.

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Appendix F5: What if gross prices respond to ACP?

- The ACP counterfactual holds gross posted prices fixed by design: it isolates the **demand-expansion channel** of the statutory discount
- The event study shows equilibrium repricing favors consumers: prices 7–10% lower in high-eligibility states after launch
- Allowing repricing would therefore **add** consumer gains; the structural CS estimates are conservative on this margin
- Read the counterfactual as the welfare effect of the discount plus induced demand and entry responses, not a decomposition of posted-price adjustment
- The firm's per-subscriber margin is unchanged because the government reimburses at the gross price

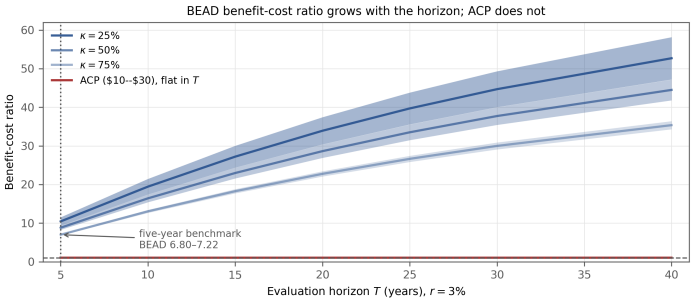
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Appendix F6: Why average HHI rises under BEAD

- At the 75% subsidy, average tract HHI moves from [4,925; 5,199] to [5,493; 5,603]
- Pure **composition effect**: BEAD serves 6,128–8,054 previously unserved tracts, and a tract served by one firm has an HHI of 10,000
- Adding newly served monopoly tracts raises the cross-sectional mean even as concentration falls *within* already-served tracts
- Consumer surplus in each newly served tract rises from zero, so welfare rises despite the index
- In this context the index measures expanded access, not increased market power

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Appendix F7: The BEAD ratio scales with the horizon



- The cost is paid once and is not discounted; only the welfare stream is, so the BCR **scales with the annuity factor**
- Across rates 1–7% at five years, every BEAD lower bound stays **above 6** and NPV stays positive ($74.6–$118.2B at 75%)
- The *sign* is robust; the *magnitude* is not. Quote the horizon with the number
- ACP's ratio is invariant to r and T; any non-trivial infra-marginal share pushes its efficiency-adjusted BCR below one

Horizon T (years), $\kappa = 75\%$, $r = 3\%$	5 (benchmark)	10	15	25	40
BEAD BCR	6.80–7.22	12.67–13.44	17.73–18.81	25.85–27.44	34.32–36.42

Appendix F8: The sequencing logic, and one honest gap

- Unserved, high-cost markets: infrastructure absence binds; fixed-cost relief delivers large, durable gains *Model implication*
- In served low-income markets, the binding constraint is price: an affordability program raises take-up and surplus at roughly break-even fiscal efficiency *Model implication*
- Where BEAD induces entry, it creates the supply-side conditions (more competing products) that raise the efficiency of a subsequent demand subsidy
- Reconciles with Kearns (2026, working paper): in already-served Seattle the demand-side tool wins, exactly the condition under which this framework also favors it

One honest gap, and it is BEAD-specific

Duopoly tracts gain only \$2.0–\$3.1 per household from BEAD: once one competitor exists, a fixed-cost discount adds little, and a subsidized third entrant needs more than 75%. ACP does reach these tracts (\$6.0–\$6.1, its largest stratum), so the two instruments are not both weak in the same place.

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Appendix F9: Distributional effects, full table

Market type	Baseline CS [$/hh/mo]	BEAD ΔCS [$/hh/mo]	ACP ΔCS [$/hh/mo]
By median household income			
Q1 (lowest, ≤$44k)	87.5–89.9	+16.7–20.7	+8.4–8.6
Q2 ($44k–$59k)	109.0–111.8	+12.4–15.4	+5.9–6.2
Q3 ($59k–$81k)	124.7–127.9	+9.8–13.0	+4.1–4.3
Q4 (highest, >$81k)	133.9–138.5	+9.9–14.4	+2.2–2.4
By number of active providers			
Monopoly	76.5–86.5	+41.5–51.6	+3.2–3.5
Duopoly	132.4–133.3	+2.0–3.1	+6.0–6.1
3+ providers	124.8–125.9	+2.5–3.8	+4.9–5.2
By tract population			
Q1 (lowest, ≤2,953)	99.9–105.5	+10.7–19.4	+6.9–7.2
Q4 (highest, >5,581)	124.9–128.3	+9.7–12.9	+2.8–3.0

75% BEAD subsidy; $30 ACP discount. $[Q_{2.5}, Q_{97.5}]$ across 50 fixed-cost draws, differenced draw-by-draw against the same draw's baseline. Monopoly tracts start from the *lowest* baseline CS and gain the most from BEAD. Panel B conditions on ≥ 1 baseline provider, so it excludes the wholly unserved tracts BEAD also reaches.

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Appendix G1: External validity and scope

- Estimates are long-run structural returns conditional on the modeled entry response; they exclude administrative and implementation costs
- The 2020 baseline predates both programs; post-2020 cost shocks (construction inflation, labor) would shift levels but not the ordering of margins
- The complementarity is a property of the deployment technology (costly to build, cheap to use), not of these two programs: the same margins recur in electricity, water, transportation, and payments
- The Seattle contrast (Kearns, 2026) is the right external check: where deployment gaps are closed, the demand-side tool wins, exactly as this framework predicts

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Appendix G2: Limitations in full

Measurement and data

- URS prices are urban: BEAD reduced-form effects are lower bounds; rural price levels extrapolated
- Market shares inferred from adoption bins, not observed; validated nationally within 1%
- 9.6% negative marginal costs; excluded and documented

Model and extrapolation

- ACP counterfactual holds gross prices fixed: a lower bound
- Wide fixed-cost bounds: partial identification by design
- The BEAD BCR scales with the horizon (6.80–7.22 at five years; 25.85–27.44 at 25); the sign never changes
- Spillovers and administrative costs excluded on both sides

None of these reverses the ordering: BEAD's five-year floor of 6.80 and ACP's near-break-even hold at every rate and horizon I compute.

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